

School of Computer Science and Engineering

Project Synopsis

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Title of the Project	Real-Time Sign Language Recognition System Using Computer Vision and Embedded Display Module		
Objective/ Purpose	1.To Develop a Real-Time Sign Language Recognition System 2.To Integrate AI with Embedded Hardware 3.To Provide a Low-Cost Assistive Communication Solution		
Abstract	Communication barriers between hearing-impaired individuals and the general public remain a significant challenge in society. This project presents a real-time Sign Language Recognition System using computer vision and embedded hardware integration. The system uses a webcam to capture hand gestures and processes them using machine learning algorithms for gesture classification. MediaPipe and image processing techniques are used to detect and track hand landmarks accurately. The recognized gestures are converted into corresponding text outputs. The interpreted text is then transmitted to an ESP32 microcontroller. A Nokia 5110 LCD display module is used as an embedded hardware output device. This ensures the inclusion of a dedicated external communication interface apart from the laptop. The system is designed to be low-cost, portable, and efficient. The proposed solution aims to improve communication accessibility for hearing-impaired individuals in everyday situations.		

Hardware & Software Requirements	Hardware Requirements 1. Webcam – To capture real-time hand gesture input. 2. ESP32 Microcontroller – To receive processed data from the system and control the display module. 3. Nokia 5110 (PCD8544) LCD Display – To display the recognized text output. 4. Breadboard – For circuit connections and prototyping. 5. Jumper Wires – To connect ESP32 and LCD module. 6. USB Cable – For programming and serial communication. 7. Power Supply / Battery (Optional) – For standalone hardware operation. 8. Buzzer (Optional) – For audio feedback indication.	Software Requirements 1. Python – Programming language for AI and computer vision implementation. 2. OpenCV – For image processing and webcam handling. 3. MediaPipe – For hand landmark detection and tracking. 4. Scikit-learn / TensorFlow – For gesture classification model training. 5. Arduino IDE – For programming the ESP32 microcontroller. 6. Serial Communication Library (PySerial) – For transmitting data from laptop to ESP32. 7. Operating System (Windows/Linux/macOS) – For development and execution.
Innovation/Novelty involved	1. The project combines real-time computer vision-based sign language recognition with an external embedded hardware display, instead of showing output only on a laptop screen. 2. It provides a low-cost and practical assistive communication solution using easily available components like ESP32 and Nokia 5110 display. 3. The system is designed to be expandable, allowing future integration with IoT, mobile applications, or smart home control.	

Algorithms / Technology Planned	1. MediaPipe Hand Tracking Algorithm – Used to detect and extract real-time hand landmarks from webcam input for accurate gesture analysis. 2. Machine Learning Classification (KNN / Random Forest) – Used to classify hand gestures based on extracted landmark features. 3. OpenCV Image Processing – Used for video capture, frame processing, and real-time visualisation. 4. Serial Communication Protocol – Used to transmit recognised text from the Python system to the ESP32 microcontroller. 5. Embedded Display Control (SPI Protocol) – Used to interface ESP32 with the Nokia 5110 LCD for displaying output text.
Methodology	● Capture real-time hand gesture input using a webcam. ● Use MediaPipe to detect and extract hand landmarks (finger joint coordinates). ● Preprocess the landmark data and convert it into feature vectors for classification. ● Apply a machine learning algorithm (KNN / Random Forest) to recognize and classify the gesture. ● Convert the recognized gesture into corresponding text output. ● Transmit the recognized text to the ESP32 microcontroller using serial communication. ● Display the interpreted text on the Nokia 5110 LCD module.

Expected Outcome s	● Successful real-time detection and recognition of predefined sign language gestures using computer vision techniques. ● Accurate conversion of recognized gestures into corresponding text output. ● Seamless transmission of recognized text from the laptop to the ESP32 microcontroller via serial communication. ● Proper display of interpreted text on the Nokia 5110 embedded display module. ● Development of a low-cost, functional assistive communication prototype integrating AI and embedded hardware. ● Demonstration of an expandable system that can be further enhanced with speech output or IoT features.
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Signature of the Guide

Project Coordinator